NAME ____________ DATE ____________

1
COMP

Functions and functional zones

You should be able to: State the functions of the respiratory system and distinguish the conducting zone from the respiratory zone by structure and by role in gas exchange.

PROMPT ☐ A ☐ B PASS 1 COLOUR ________ PASS 2 COLOUR ________ ALSO A LAB SKILL

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boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

NAME ____________ DATE ____________

2

COMP

Pressure gradients and airflow

You should be able to: Apply Boyle's law to inspiration and expiration and relate alveolar pressure to atmospheric pressure at each point in a quiet breath.

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NAME ______________ DATE ______________

3

COMP

Intrapleural pressure

You should be able to: Explain why intrapleural pressure is subatmospheric and predict what happens to the lung and chest wall when the pleural seal is broken.

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NAME ____________ DATE ____________

4
COMP

Compliance and elastic recoil

You should be able to: Define lung compliance and elastic recoil and predict the effect of fibrosis and of emphysema on each.

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5

COMP

Surfactant and surface tension

You should be able to: Explain how surfactant lowers surface tension and why its absence causes alveolar collapse in the premature newborn.

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6
COMP

Airway resistance

You should be able to: Identify the main determinant of airway resistance and predict the effect of bronchoconstriction, mucus, and sympathetic stimulation on airflow.

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NAME ____________ DATE ____________

7

COMP

Lung volumes and capacities

You should be able to: Define tidal volume, inspiratory and expiratory reserve volume, residual volume, vital capacity, and total lung capacity and read each from a spirogram.

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8

COMP

Obstructive and restrictive patterns

You should be able to: Use the ratio of forced expiratory volume in one second to forced vital capacity to classify a spirometry result as obstructive or restrictive.

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 NAME ______________ DATE ____________

9

COMP

Alveolar ventilation and dead space

You should be able to: Calculate minute ventilation and alveolar ventilation and explain why slow deep breathing ventilates the alveoli better than rapid shallow breathing.

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10
COMP

Spirometry

You should be able to: Record a spirogram, calculate the lung volumes and capacities available from it, and compare the results with predicted values.

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11

COMP

Partial pressures

You should be able to: Apply Dalton's law to calculate the partial pressure of oxygen in inspired and alveolar air and state the normal partial pressures in alveolar air, arterial blood, and venous blood.

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NAME ______________________ DATE ______________

12

COMP

Diffusion at the respiratory membrane

You should be able to: Apply the determinants of diffusion to the respiratory membrane and predict the effect of edema, fibrosis, and emphysema on gas transfer.

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13

COMP

Ventilation perfusion matching

You should be able to: Explain how local hypoxic vasoconstriction and bronchiolar responses match ventilation to perfusion and predict the consequence of a mismatch.

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14

COMP

Oxygen transport and the oxyhemoglobin curve

You should be able to: Explain the sigmoid shape of the oxyhemoglobin dissociation curve and state the physiological advantage of the steep and the flat portions.

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15

COMP

Curve shifts

You should be able to: Predict the direction of the oxyhemoglobin curve shift produced by a change in pH, carbon dioxide, temperature, or 2,3 bisphosphoglycerate and state what the shift means for tissue oxygen delivery.

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16

COMP

Carbon dioxide transport

You should be able to: Name the three forms in which carbon dioxide is carried, state the approximate percentage of each, and trace the chloride shift in a systemic capillary.

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17

COMP

Bohr and Haldane effects

You should be able to: Distinguish the Bohr effect from the Haldane effect and explain how the two work together at the tissue and at the lung.

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18

COMP

Oxygen content versus partial pressure

You should be able to: Distinguish oxygen content from oxygen partial pressure and saturation and explain why anemia and carbon monoxide poisoning reduce delivery differently.

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19

COMP

Respiratory centers

You should be able to: Identify the medullary and pontine respiratory centers and explain how the basic breathing rhythm is generated and modified.

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20

COMP

Chemoreceptor control

You should be able to: Compare central and peripheral chemoreceptors by stimulus and explain why carbon dioxide is the primary minute to minute drive to breathe.

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21
COMP

Ventilation in exercise and at altitude

You should be able to: Predict the ventilatory response to exercise and to acute and chronic altitude exposure and explain the acclimatization changes.

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22

COMP

Ventilatory response testing

You should be able to: Measure the change in breathing rate and depth after hyperventilation, breath holding, and rebreathing and explain each result by the change in carbon dioxide.

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23
COMP

Buffer systems

You should be able to: Compare the bicarbonate, phosphate, and protein buffer systems by location and speed and explain why bicarbonate is the dominant extracellular buffer.

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24

COMP

Respiratory control of pH

You should be able to: Explain how a change in ventilation shifts the carbon dioxide and bicarbonate equilibrium and therefore plasma pH.

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